EXPERIMENT NO:02

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    'Sky': ['Sunny', 'Sunny', 'Rainy', 'Sunny'],
    'AirTemp': ['Warm', 'Warm', 'Cold', 'Warm'],
    'Humidity': ['Normal', 'High', 'High', 'High'],
    'Wind': ['Strong', 'Strong', 'Strong', 'Strong'],
    'Water': ['Warm', 'Warm', 'Warm', 'Cool'],
    'Forecast': ['Same', 'Same', 'Change', 'Change'],
    'EnjoySport': ['Yes', 'Yes', 'No', 'Yes']
}

df = pd.DataFrame(data)
attributes = list(df.columns[:-1])

S = ['0'] * len(attributes)
G = [['?'] * len(attributes)]

for _, row in df.iterrows():
    x = list(row[attributes])

    if row['EnjoySport'] == 'Yes':
        G = [g for g in G if all(
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            a == '?' or a == v for a, v in zip(g, x)
        )]

        for i in range(len(S)):
            if S[i] == '0':
                S[i] = x[i]
            elif S[i] != x[i]:
                S[i] = '?'
    else:
        new_G = []
        for g in G:
            if all(a == '?' or a == v for a, v in zip(g, x)):
                for i in range(len(g)):
                    if g[i] == '?' and S[i] not in ('0', x[i]):
                        h = g.copy()
                        h[i] = S[i]
                        new_G.append(h)
            else:
                new_G.append(g)
        G = new_G

print("Specific Boundary (S):", S)
print("General Boundary (G):", G)

# Graph 1: Hypothesis boundaries
plt.figure(figsize=(10, 5))
```

```
plt.plot(attributes, [1 if x != '?' else 0 for x in S],
         marker='o', label='Specific Boundary S')

plt.plot(attributes, [1 if x == '?' else 0 for x in S],
         marker='s', label='Generalized attributes in S')

plt.title("Candidate-Elimination: Specific Boundary")
plt.xlabel("Attributes")
plt.ylabel("Value (1 = True, 0 = False)")
plt.ylim(-0.2, 1.2)
plt.grid(True)
plt.legend()
plt.xticks(rotation=30)
plt.tight_layout()
plt.show()

# Graph 2: Display hypothesis table
plt.figure(figsize=(12, 3))
plt.axis('off')
plt.title("Candidate-Elimination Hypothesis Boundaries")

table_data = [['S'] + S] + [['G'] + g for g in G]

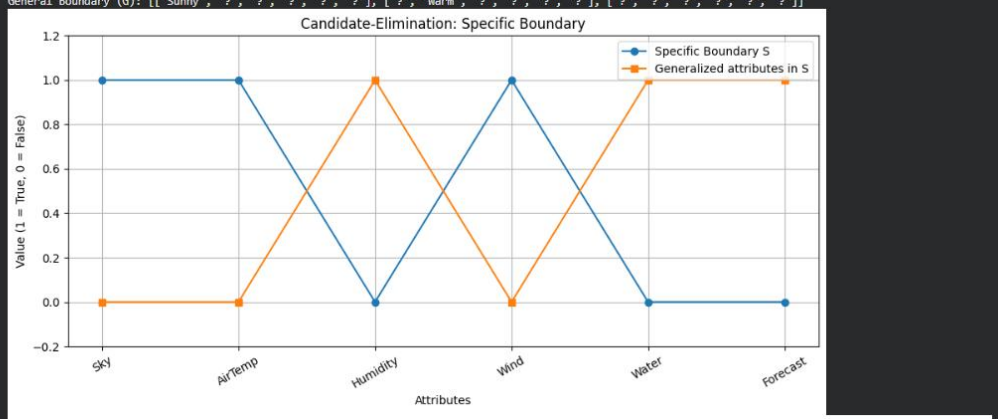
plt.table(
    cellText=table_data,
```

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    colLabels=['Boundary'] + attributes,
    cellLoc='center',
    loc='center'
)

plt.tight_layout()
plt.show()
```

OUTPUT:

```
Specific Boundary (S): ['Sunny', 'Warm', '?', 'Strong', '?', '?']
General Boundary (G): [['Sunny', '?', '?', '?', '?', '?'], ['?', 'Warm', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?']]
```



Candidate-Elimination Hypothesis Boundaries

Boundary	Sky	AirTemp	Humidity	Wind	Water	Forecast
S	Sunny	Warm	?	Strong	?	?
G	Sunny	?	?	?	?	?
G	?	Warm	?	?	?	?
G	?	?	?	?	?	?